

SmartPark Optimization: A Mathematical Programming Approach for Urban Parking Allocation in Brescia

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1. Introduction & Motivation

1.1 The Parking Problem in Italian Cities

Urban parking management represents one of the most relevant challenges for modern cities, especially in densely populated urban areas characterized by high traffic congestion and limited available space. In many Italian cities, the increasing number of private vehicles, combined with historical urban structures and constrained road networks, has generated significant pressure on parking infrastructure. Drivers frequently experience difficulties in finding available parking spaces, particularly near commercial districts, business centres, universities, and transportation hubs.

The inefficiency of parking systems produces several negative effects on urban mobility and quality of life. A considerable portion of urban traffic is generated by vehicles searching for parking spaces, leading to additional congestion, longer travel times, increased fuel consumption, and higher levels of air pollution. Furthermore, the mismatch between parking demand and parking supply may reduce the accessibility and attractiveness of specific urban areas.

According to ACI 2024 data, in Italy the national car ownership rate stands at around 670 vehicles per 1,000 inhabitants, with greater peaks in medium sized cities like Brescia. This high number of vehicles generates a demand that exceeds the availability supply in central neighbourhoods, leading to congestion, CO₂ emissions linked to the search for a free space and diseconomies for businesses and residents.

This type of problem is extendable in other countries, and has even been the subject of European studies, that estimate that up to 30% of urban traffic during peak hours is generated by vehicles searching for parking spaces, with an average time of searching of over 8 minutes. The overall social cost of this inefficiency, in terms of time lost, pollution and fuel, has been quantified at around €900 per inhabitant per year in major European cities, according to the INRIX 2023 report on road congestion.

For these reasons, urban parking planning has become an important topic within the fields of operations research, transportation systems, and smart city management. Optimization models can support public administrations in identifying efficient parking allocation strategies, evaluating potential investments, and improving the overall accessibility of the urban environment while balancing economic and social objectives.

1.2 The Case of Brescia and Motivation for the Study

This project focuses on the city of Brescia, a medium-sized urban area in Northern Italy characterized by a heterogeneous distribution of residential zones, commercial activities, university facilities, and business districts. Like many Italian cities, Brescia experiences significant parking pressure in the most active areas of the city, particularly during peak hours.

The choice of Brescia as the case study is motivated both by its relevance as an urban mobility context and by our direct familiarity with the city. As students of the University of Brescia, the urban dynamics, traffic conditions, and parking issues of Brescia represent a concrete and observable problem that can be analyzed through quantitative optimization techniques.

The project considers a network composed of:

- 20 demand nodes, representing the most crowded and strategically relevant areas of the city;
- 26 existing parking facilities already operating within the urban area;
- 5 additional candidate parking sites that may be activated depending on the ability of the current system to satisfy parking demand.

The model also incorporates different categories of parking users and temporal demand variations. This structure allows the model to better capture the variability of urban parking usage and to represent more realistic mobility patterns.

The main objective of this project is to develop an optimization model capable of supporting strategic parking allocation decisions within the city of Brescia. The proposed model aims to evaluate whether the existing parking infrastructure is sufficient to satisfy urban demand or whether additional parking facilities should be activated.

The project aims to answer the following research questions:

1. Is the current parking infrastructure sufficient to satisfy the parking demand generated by the main urban areas of Brescia?
2. Which candidate parking facilities should be activated in order to improve demand coverage and system efficiency?
3. How does parking allocation vary across different user categories and time periods?
4. What is the impact of pricing policies and user behavior on parking allocation decisions?
5. How do different optimization objectives influence the final configuration of the parking system?

2. Literature Review

2.1 Facility Location Problems: An Overview

Facility Location Problems (FLPs) represent one of the most widely studied classes of optimization problems within operations research and urban planning. Their general objective is to determine the optimal location and allocation of facilities in order to efficiently satisfy spatially distributed demand while minimizing costs or maximizing system performance.

Typical applications include logistics networks, warehouses, hospitals, public services, transportation hubs, and urban infrastructure systems. In the context of urban mobility, parking facilities can also be interpreted as service facilities that must satisfy geographically distributed users under capacity constraints.

Among the different variants of FLPs, the Capacitated Facility Location Problem (CFLP) is particularly relevant to this project. In a CFLP, facilities have limited capacities and demand must be allocated across open facilities while respecting operational constraints. These problems are commonly formulated as Mixed-Integer Linear Programming (MILP) models, where binary variables determine whether facilities are activated and continuous variables represent demand allocation decisions.

Our project belongs to this family of optimization problems. More specifically, the proposed framework can be interpreted as a capacitated facility location and demand allocation model applied to urban parking management.

2.2 Optimization Models for Urban Parking Systems

Urban parking optimization has received increasing attention in recent years due to the growing importance of sustainable mobility, congestion reduction, and smart city planning. Existing literature generally focuses on improving the efficiency of parking systems through optimization techniques, simulation models, pricing mechanisms, or intelligent transportation systems.

Several studies model parking allocation as a facility location problem in which users are assigned to parking lots according to distance, availability, or travel time. A common objective is the minimization of walking distance and traffic congestion generated by vehicles searching for parking spaces.

Our model includes:

- multiple parking facilities with capacity constraints,
- geographically distributed demand nodes,
- distance-sensitive user allocation,
- heterogeneous user categories,
- time-dependent parking demand,
- strategic activation decisions for new parking sites.

2.3 Positioning of the Proposed Model

The proposed framework explicitly differentiates users according to behavioural and economic characteristics. Residents, workers, and visitors are associated with different sensitivities to walking

distance, different unmet demand penalties, and different tariff structures. This allows the model to represent heterogeneous urban mobility behaviour in a more realistic way.

Another important feature of the model is the temporal segmentation of demand into morning, afternoon, and night periods. Parking usage patterns vary significantly throughout the day, especially in mixed urban environments where residential, commercial, university, and healthcare activities coexist. Introducing multiple time periods enables the model to capture these variations more accurately.

The project introduces a multi-scenario perspective through the comparison of two alternative objective functions:

- cost minimization,
- social profit maximization.

This comparison constitutes one of the main analytical contributions of the project. Many studies focus exclusively on minimizing operational costs or maximizing revenues, while the SmartPark framework explicitly investigates the potential tension between economic efficiency and social welfare. The comparison between the two scenarios allows the analysis to evaluate whether the economically optimal parking configuration also corresponds to the socially preferable solution.

Finally, the model is strongly application-oriented and specifically calibrated for the urban context of Brescia. Demand nodes, parking facilities, tariff structures, and behavioural parameters are designed to reflect the characteristics of the local urban environment rather than relying on purely theoretical assumptions.

3. Data Collection and Preparation

3.1 Set I – Parking Sites

Set I comprises all parking facilities considered in the model, partitioned into two subsets: existing open parking lots and candidate sites for potential new construction. The complete set consists of 31 elements.

The 26 existing sites correspond to the paid parking facilities currently operated in the city of Brescia, as documented in the inventory published by Brescia Mobilità and in the official data made available by the Municipality. For each site, the capacity C_i was taken directly from the same source, which lists the total number of spots. Because these facilities are already operational, their construction cost is treated as a sunk cost and the corresponding fixed cost parameter is set to zero ($F_i = 0$). In the model, the binary variable X_i is fixed to 1 for all of these sites, meaning they are always active and contribute their full capacity to the allocation problem.

The five candidate sites were identified through a three-step process. First, a geographic analysis of the distance matrix revealed that several demand nodes in the southern and western areas of the city were poorly served by existing facilities, as no parking lot was located within a reasonable walking distance. Second, areas with chronically high parking demand and documented saturation (in particular the hospital district in the north and the Lamarmora neighbourhood in the south) were prioritised. Third, the feasibility of each candidate was verified against the Piano di Governo del Territorio (PGT) published by the

Municipality of Brescia, ensuring that the selected locations correspond to zones where new construction is legally permissible.

For candidate sites, the fixed cost F_i represents the total estimated construction cost, derived from standard unit cost benchmarks for Italian urban parking structures: approximately €10,000 per spot for multi-storey facilities and €4,000–5,000 per spot for surface or Park-and-Ride layouts. To make F_i comparable with the daily revenue and penalty terms in the objective functions, it is annualised over a 20-year useful life ($F_i / 20$).

ID	Name	Address	Capacity (spots)	Fixed cost F_i	Status
I01	Camper Poliambulanza	Via Morelli	10	—	Existing
I02	Crystal Superficie	Via Aldo Moro 17	50	—	Existing
I03	Apollonio	Via Apollonio, 15	115	—	Existing
I04	Castellini	Via Castellini / Via Mantova	215	—	Existing
I05	Ospedale Nord (surface)	P.za S. Padre Pio, 1	150	—	Existing
I06	Stadio Rigamonti	Via dello Stadio	496	—	Existing
I07	Poliambulanza	Via Morelli	350	—	Existing
I08	Prealpino	Via Triumplina	1005	—	Existing
I09	Autosilo Uno	V.le Vittorio Emanuele II, 3	336	—	Existing
I10	S. Eufemia-Buffalora	Via Agostino Chiappa	398	—	Existing
I11	Casazza	Via Triumplina 181/02	160	—	Existing
I12	Arnaldo Park	Piazzale Arnaldo	300	—	Existing
I13	Crystal (multi-storey)	Via Aldo Moro 17	400	—	Existing
I14	Palagiustizia	Via Gambara, 44	600	—	Existing
I15	Stazione	V.le Stazione, 51	1000	—	Existing
I16	Benedetto Croce	P.tta Don L. Sturzo	72	—	Existing
I17	San Domenico	Piazza San Domenico	72	—	Existing
I18	Freccia Rossa	Viale Italia, 31	2500	—	Existing

I19	Piazza Mercato	Piazza del Mercato	190	—	Existing
I20	Randaccio	Via Lupi di Toscana, 4	180	—	Existing
I21	Fossa Bagni	Piazza Fossa Bagni	560	—	Existing
I22	D'Azeglio	Via M. D'Azeglio, 4a	35	—	Existing
I23	Ospedale Sud	Via Ducco / Via dal Monte 44	500	—	Existing
I24	Ospedale Nord (multi-storey)	P.za S. Padre Pio, 1	1260	—	Existing
I25	Goito	Via Spalto S. Marco, 8	215	—	Existing
I26	Piazza Vittoria	Piazza della Vittoria	520	—	Existing
I27	HUB Lamarmora Sud	Via Malta / ex-Zooprofilattico	450	€ 4,500,000	Candidate
I28	Parcheggio Sud Stazione	Via Dalmazia / Via Sostegno	350	€ 3,500,000	Candidate
I29	Ospedale Nord-Ovest	Via Nikolajewka	300	€ 1,200,000	Candidate
I30	Parcheggio ex Pietra	Via Dalmazia / Via Corsica	450	€ 4,500,000	Candidate
I31	Mega Park&Ride Sud	Via Orzinuovi / Chiesanuova	950	€ 4,750,000	Candidate

Table 3.1 — Set I: complete list of parking sites.

3.2 Set J — Demand nodes

Set J represents the spatial demand side of the model: each node is a geographic point that generates parking demand from one or more user types. The set contains 20 nodes, covering the main attractors, institutional facilities, transport hubs, and residential neighbourhoods of Brescia.

Node selection followed a top-down logic. Starting from the full list of existing parking lots and from the locations identified as parking-scarce in the city's mobility plans, we defined demand nodes that either concentrate a significant number of users of a specific type (e.g. Spedali Civili for workers, Santa Giulia for visitors), or correspond to densely populated residential areas with documented overnight parking pressure (e.g. Quartiere Carmine, Mompiano). Nodes were distributed across five functional categories: historic centre, attractors, universities, hospitals, and residential areas.

ID	Name	Category	Dominant user type	Justification
J01	Piazza della Vittoria	Historic centre	Visitors / Workers	Main civic and commercial hub
J02	Piazza Paolo VI / Duomo	Historic centre	Visitors	Major tourist attraction
J03	Teatro Grande	Historic centre	Visitors	Cultural venue, Corso Zanardelli
J04	Palazzo di Giustizia	Historic centre	Workers	Law courts + legal offices district
J05	Santa Giulia / Capitolium	Attractor	Visitors	UNESCO museum complex
J06	Castello di Brescia	Attractor	Visitors	Hilltop park and museum
J07	Piazzale Arnaldo	Attractor	Visitors / Residents	Nightlife, market, restaurants
J08	Stadio Rigamonti	Attractor	Visitors	Average of non-match days
J09	Brixia Forum	Attractor	Visitors / Workers	Exhibition and congress centre
J10	Università – central campus	University	Workers	UniBS staff + commuting students
J11	Università – Engineering campus	University	Workers	Northern campus + research spin-offs
J12	Spedali Civili	Hospital	Workers	staff + patients / families
J13	Poliambulanza	Hospital	Workers	Private hospital + commuters
J14	Ist. Clinico Città di Brescia	Hospital	Workers / Residents	Eastern hospital complex

J15	Quartiere Carmine	Residential	Residents	Dense historic residential area
J16	Zona Stazione / Fiumicello	Residential	Residents / Workers	Station hub, hotels, offices
J17	Mompiano / Prealpino	Residential	Residents	Northern hillside neighbourhoods
J18	Urago Mella / Sanpolino	Residential	Residents	Eastern peripheral residential area
J19	Lamarmora / Brescia Due	Residential + Tertiary	Workers / Residents	12,000 residents + tertiary pole
J20	San Polo / Villaggio Sereno	Residential	Residents	Southern residential area

Table 3.2 — Set J: complete list of demand nodes.

3.3 Set K — User types

Set K distinguishes three behavioural profiles that differ in parking duration, price sensitivity, distance tolerance, and social priority. This differentiation is essential to capture the trade-offs between the two objective functions: a cost-minimising approach tends to prioritise users with high unmet-demand penalties (residents), while a social-profit-maximising approach is drawn towards users who generate higher tariff revenues (visitors and workers with long stays).

Residents are defined as individuals parking their private vehicle near their home address, predominantly during the night period. They are assigned the lowest tariff ($P_{\text{night}} = \text{€}0.18/\text{h}$ at night, reflecting the Brescia Mobilità Park City Card monthly subscription of €80 converted to an hourly equivalent), the highest distance penalty α (residents are least tolerant of walking, especially at night), and the highest unmet-demand penalty β (not finding a parking space near home constitutes a severe social disservice).

Workers are daily commuters arriving by car to reach their workplace. They are assigned an intermediate tariff (€1.30/h, corresponding to the fourth-to-sixth-hour rate charged by Brescia Mobilità multi-storey facilities), a medium distance sensitivity, and an intermediate unmet-demand penalty. The dominant period is the morning shift (07:00–14:00), when 70% of their daily demand is concentrated.

Visitors encompass tourists, cultural facility users, hospital visitors, and short-term shoppers. They are assigned the highest tariff (up to €2.00/h in the afternoon, averaging zone A street rates and multi-storey rates), the lowest distance penalty (a short walk is considered acceptable as part of the visit experience), and the lowest unmet-demand penalty (visitors generally have more flexible alternatives). Their demand peaks in the afternoon period.

Type k	Description	Tariff level	α sensitivity	Notes
Residents	Local residents parking near home	Low (Park City Card –50%)	High	High β penalty; dominated night period
Workers	Employees commuting daily by car	Medium (daily / monthly pass)	Medium	Long stay (avg. 7 h); dominant in morning period
Visitors	Short-term users, tourists, patients	High (full hourly rate)	Low	Short stay (avg. 2.5 h); dominant in afternoon

Table 3.3 — Set K: user type

3.4 Set T — Time periods

Set T partitions the 24-hour day into three non-overlapping periods of unequal length. The choice of three periods reflects a deliberate trade-off between model complexity and analytical richness. Three periods proved sufficient to capture the three structurally different demand profiles.

The morning period (07:00–14:00) was defined to capture the arrival of workers at shift start (07:00–09:00) and their occupancy through the midday hours. The afternoon period (14:00–21:00) covers the second work shift, commercial activity, cultural visits, and the early evening social events. The cut-off at 21:00 coincides with the transition from active daytime tariffs to reduced evening/night rates in most Brescia Mobilità facilities, and with the start of residential overnight parking. The night period (21:00–07:00) runs for ten hours and is dominated by residential demand.

Period t	Time window	Dominant users	Demand level	Notes
Morning	07:00 – 14:00	Workers	High	Peak arrival of shift workers
Afternoon	14:00 – 21:00	Visitors	Medium–high	Commercial, cultural and leisure activities
Night	21:00 – 07:00	Residents	Low	Residential overnight parking; high β if unmet

Table 3.4 — Set T: time period definitions

3.5 Distance Matrix d_{ij}

The distance matrix d_{ij} represents the estimated walking distance, expressed in metres, between each parking site i and each demand node j . The matrix includes all possible combinations between the 31 parking sites and the 20 demand nodes considered in the model.

This parameter plays an important role in the optimisation process because distance directly affects the allocation of users to parking facilities. In particular, longer walking distances generate higher penalty costs in the objective functions, making nearby parking lots more attractive whenever sufficient capacity is available.

The distances were estimated using a simplified geographic approach based on three main steps.

$$d(i, j) = 2R * \arcsin \left(\sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) * \cos(\phi_2) * \sin^2\left(\frac{\Delta\lambda}{2}\right)} \right)$$

First, the geographic coordinates (latitude and longitude) of each parking site and demand node were collected using Google Maps and verified through the official Brescia Mobilità parking inventory.

Second, the straight-line distance between each pair was calculated using the Haversine formula, which estimates the geodetic distance between two points on the Earth's surface based on their coordinates.

Finally, since pedestrians cannot move in a perfectly straight line within an urban environment, the resulting distances were multiplied by a correction factor equal to 1.3. This adjustment approximates the actual walking distance by accounting for road layouts, intersections, buildings, and irregular street patterns typical of Italian cities such as Brescia.

The final distances were rounded to the nearest metre and stored as input parameters for the optimisation model.

The resulting matrix contains 620 origin–destination pairs and reflects the geographic structure of the city. Short distances are mainly observed between parking lots and demand nodes located in the historic centre, while peripheral residential areas and park-and-ride facilities generate longer distances.

The matrix also highlights the strategic positioning of the proposed candidate parking sites. Each new parking facility was intentionally located close to at least one underserved demand area, allowing the optimisation model to evaluate whether opening the site can improve demand coverage and reduce congestion in critical urban zones.

ID	Parcheggio	J01 Piazza della Vittoria	J02 Piazza Paolo VI / Duomo	J03 Teatro Grande	J04 Palazzo di Giustizia	J05 Santa Giulia / Capitoli	J06 Castello di Brescia	J07 Piazzale Arnaldo	J08 Stadio Rigamon- ti	J09 Brixia Forum	J10 Universi- tà sede centrale	J11 Universi- tà Ingeg- nerie	J12 Spedali Civili	J13 Poliamb- ulanza	J14 Ist. Clinico Città BS	J15 Quartier e Carmine	J16 Zona Stazione Fiumice	J17 Mompia- no/Preal- pino	J18 Urago Mella/Sa- npolino	J19 Lamarm- ora / Brescia	J20 San Polo / Villaggio
I01	Camper Poliambulanza	2989	2940	2973	3630	2875	3313	2281	3909	4153	2574	6239	5085	48	3195	3613	2928	7468	2390	1676	1659
I02	Crystal Superficie	332	470	244	921	1116	1188	1078	1941	2228	902	4331	3253	2775	2618	957	780	5006	4388	2360	4388
I03	Apollonio	1291	1409	1130	1624	1819	2100	1679	2859	2048	1336	5277	4180	2228	3209	1895	755	5930	4252	1421	3804
I04	Castellini	2173	2163	2125	2760	2082	2649	1662	3328	3273	1847	5743	4590	946	2995	2811	2024	8795	3100	1168	2541
I05	Ospedale Nord (sup.)	2976	2813	3194	3199	2556	2111	2939	1354	5008	2913	1113	53	5176	2274	2582	3952	2890	5392	5506	6543
I06	Stadio Rigamonti	1651	1471	1877	2016	1250	759	1646	0	3839	1562	2443	1321	3956	1658	1377	2647	3665	4601	4157	5423
I07	Poliambulanza	3036	2988	3019	3676	2722	3360	2329	2956	4187	2622	8284	5130	0	3234	3990	2969	7546	2380	1689	1613
I08	Prealpino	5085	5068	5192	4723	5318	4656	5696	4178	5907	5398	2901	3515	8008	5847	4448	5995	583	8739	7725	9550
I09	Autosilo Uno	276	505	77	606	1278	1177	1320	1894	2073	810	4218	3176	3088	2772	723	754	4765	4687	2586	4700
I10	S.Eufemia-Bufferalora	4880	4818	5084	5751	4143	4765	3903	4971	6800	4406	6606	5824	2874	3425	5446	5379	8519	506	4562	2465
I11	Casazza	3141	3115	3261	2848	3380	2714	3753	2311	4275	3442	2136	2062	6057	3898	2509	3818	1546	6912	5806	7608
I12	Arnaldo Park	1098	918	1248	1873	397	1040	0	1646	3290	516	4051	2898	2329	1634	1553	1803	5222	3449	2632	3855
I13	Crystal (autosilo)	332	470	244	921	1116	1188	1078	1941	2228	902	4331	3253	2775	2618	957	780	5006	4388	2360	4388
I14	Palagiustizia	789	975	705	43	1754	1419	1884	1990	1855	1369	4090	3151	3707	3177	814	1017	4341	5299	3064	5320
I15	Stazione	1003	1212	771	974	1892	1919	1803	2647	1496	1371	4958	3929	2969	3389	1418	0	5357	4885	2075	4556
I16	Benedetto Croce	495	263	661	1229	556	692	655	1446	2783	178	3881	2767	2797	2055	915	1390	4783	4101	2724	4379
I17	San Domenico	476	385	693	944	889	500	1137	1188	2707	706	3538	2476	3324	2246	430	1460	4282	4577	3137	4909
I18	Freccia Rossa	1904	2112	1671	1710	2762	2819	2620	3542	973	2239	5913	4811	3222	4241	2293	901	6021	5385	1873	4698
I19	Piazza Mercato	582	415	740	1358	523	825	516	1562	2830	0	4004	2878	2622	2018	1086	1371	4951	3946	2595	4203
I20	Randaccio	1154	1218	1216	776	1820	1255	2082	1560	2472	1623	3408	2546	4169	3028	538	1735	3828	5521	3735	5762
I21	Fossa Bagni	294	167	525	912	845	625	1029	1356	2804	553	3731	2658	3150	2281	533	1296	4477	4478	2935	4742
I22	D'Azeglio	789	681	880	1548	668	1115	448	1833	2854	308	4275	3138	2320	2074	1362	1361	5257	3734	2332	3907
I23	Ospedale Sud	2883	2880	3116	3294	2207	1967	2546	1278	5117	2661	1753	772	4626	1532	2655	3888	3713	4620	5177	5899
I24	Ospedale Nord (autosilo)	2629	2760	3149	3177	2477	2053	2858	1293	4991	2847	1202	59	5083	2164	2555	3911	3001	5281	5436	6441
I25	Golfo	981	790	1145	1753	291	905	143	1536	3218	405	3958	2806	2457	1662	1414	1740	5080	3587	2693	3992
I26	P.za Vittoria	35	240	242	749	1031	908	1124	1637	2343	908	3993	2933	3071	2512	906	1013	4648	4545	2722	4678
I27	HUB Lamarmora Sud	2714	2760	2606	3154	2832	3339	2459	4056	3193	2516	6493	5347	1212	3823	3348	2260	7394	3590	478	2491
I28	Parcheggio Sud Stazior	1728	1938	1495	1543	2596	2644	2465	3366	1014	2074	5636	4634	3175	4081	2088	726	5872	5298	1903	4679
I29	P. Ospedale Nord-Ovest	3045	2904	3250	3172	2748	2234	3141	1504	4945	3055	970	436	5422	2624	2585	3989	2506	5744	5648	6829
I30	Parcheggio ex Pietra	1481	1649	1403	722	2399	1957	2566	2370	1767	2055	4142	3347	4381	3749	1114	1533	4010	5995	3603	5992
I31	Mega Park&Ride Sud	5193	5192	5122	5712	5033	5949	4636	8277	5646	4869	8913	7460	2330	5501	5834	4837	9820	3530	2845	1389

Table 3.5 — Distance matrix $d(i,j)$

3.6 Estimation of Parking Demand D_{jkt}

The parameter D_{jkt} represents the estimated parking demand generated by demand node j , user category k , and time period t . More specifically, the table stores the expected number of vehicles requiring parking in a specific urban area, for a given type of user and during a given time interval.

This parameter constitutes one of the core inputs of the optimisation model, since it determines how parking demand is distributed across the available parking facilities. The demand matrix is therefore used both in the objective functions and in the capacity constraints of the model.

Since no complete public dataset exists for parking demand at neighbourhood level in Brescia, the values of D_{jkt} were estimated through a multi-source approximation process combining demographic data, mobility statistics, institutional reports, and behavioural assumptions.

The estimation procedure was developed separately for the three user categories considered in the model.

Resident parking demand was estimated using population data from the ISTAT permanent census datasets at sub-municipal level. The population associated with the census areas included in each demand node was aggregated and converted into potential parking demand through the local motorisation rate.

According to ISTAT and ACI statistics, the estimated motorisation rate for the city of Brescia is approximately 620 vehicles per 1,000 inhabitants. However, not all residents use their cars daily or require parking during every time period. For this reason, an additional utilisation coefficient between 30% and 35% was applied in order to estimate the effective number of vehicles searching for parking.

The temporal distribution of resident demand was calibrated according to typical urban residential behaviour:

- 10% of resident demand assigned to the morning period,
- 30% to the afternoon period,
- 60% to the night period.

This distribution reflects the fact that residential parking pressure is significantly higher during nighttime hours, when most residents return home.

Worker demand was estimated using employment and commuting data related to the main urban attractors included in the model, such as hospitals, universities, offices, and institutional buildings.

For example:

- the Spedali Civili hospital district was estimated to employ approximately 6,500 workers,
- the University of Brescia contributes approximately 1,200 employees and staff members,
- the Tribunale district contributes roughly 400 workers.

Additional information regarding commuter flows was obtained from the ISTAT commuting matrix dataset, which provides origin–destination flows for work-related mobility.

To estimate effective parking demand, a private car usage coefficient between 50% and 60% was applied, reflecting the share of workers expected to commute by car.

Worker demand was then distributed across the different time periods:

- 70% assigned to the morning period,
- 25% to the afternoon period,
- 5% to the night period.

This allocation reflects the concentration of work-related activities during standard daytime working hours.

Visitor demand was estimated using tourism and attraction-related datasets published by the Municipality of Brescia, Regione Lombardia Open Data, and Brescia Musei.

According to the 2024 municipal tourism report, the city recorded approximately 7.1 million visitors per year, corresponding to an average of roughly 594,000 visitors per month. The data also indicate that approximately 72% of tourists visiting the historic city centre are international visitors.

Additional attendance data were collected for major cultural attractions such as Santa Giulia Museum, Capitolium, and Castello di Brescia. These values were used to estimate visitor flows associated with the main tourist demand nodes included in the model.

Since not all visitors arrive by private vehicle, a parking utilisation coefficient between 40% and 50% was applied. Visitor demand was then distributed temporally as follows:

- 35% assigned to the morning period,
- 60% to the afternoon period,
- 5% to the night period.

This distribution reflects the typical concentration of tourism and leisure activities during daytime and early afternoon hours.

Matrice domanda D _{jkt} — auto/fascia · Mattina 07–14 · Pomeriggio 14–21 · Notte 21–07												
ID	Nodo J	RESIDENTI			LAVORATORI			VISITATORI			D _{j,tot}	Tipo prev.
		Mattina	Pomerig.	Notte	Mattina	Pomerig.	Notte	Mattina	Pomerig.	Notte		
J01	Piazza della Vittoria	30	89	179	578	206	41	315	540	45	2023	Visitatori
J02	Piazza Paolo VI / Duomo	16	48	95	308	110	22	394	675	56	1724	Visitatori
J03	Teatro Grande	12	36	71	192	69	14	158	270	22	844	Visitatori
J04	Palazzo di Giustizia	6	18	36	770	275	55	79	135	11	1385	Lavoratori
J05	Santa Giulia / Capitolium	4	12	24	77	28	6	236	405	34	826	Visitatori
J06	Castello di Brescia	3	9	18	38	14	3	189	324	27	625	Visitatori
J07	Piazzale Arnaldo	60	178	357	231	82	16	284	486	40	1734	Visitatori
J08	Stadio Rigamonti	79	238	476	77	28	6	79	135	11	1129	Residenti
J09	Brixia Forum	20	59	119	192	69	14	158	270	22	923	Visitatori
J10	Università sede centrale	30	89	179	770	275	55	126	216	18	1758	Lavoratori
J11	Università Ingegneria	40	119	238	693	248	50	94	162	14	1658	Lavoratori
J12	Spedali Civili	10	30	59	2502	894	179	551	945	79	5249	Lavoratori
J13	Poliambulanza	30	89	179	962	344	69	315	540	45	2573	Lavoratori
J14	Ist. Clinico Città BS	40	119	238	308	110	22	126	216	18	1197	Lavoratori
J15	Quartiere Carmine	159	476	952	192	69	14	31	54	4	1951	Residenti
J16	Zona Stazione/Fiumicello	119	357	714	578	206	41	315	540	45	2915	Residenti
J17	Mompiano / Prealpino	198	595	1190	308	110	22	31	54	4	2512	Residenti
J18	Urago Mella / Sanpolino	238	714	1429	231	82	16	24	41	3	2778	Residenti
J19	Lamarmora / Brescia Due	238	714	1429	770	275	55	63	108	9	3661	Residenti
J20	San Polo / Vill. Sereno	179	536	1072	192	69	14	31	54	4	2151	Residenti
TOTALE SISTEMA		1511	4525	9054	9969	3563	714	3599	6170	511	39616	

Table 3.6 — Estimated demand $D(j,k,t)$

3.7 Tariff Structure and Revenue Parameters P_{kt}

The parameter P_{kt} represents the parking tariff associated with user category k during time period t . In the optimization model, this parameter is used to estimate the economic revenue generated by parking demand under Objective B (social profit maximisation).

More specifically, P_{kt} represents the equivalent parking revenue generated by a single parking event for a given user category and time period. The parameter therefore captures both the hourly parking tariff and the typical duration of parking occupancy.

The estimation of parking tariffs was based primarily on official data published by Brescia Mobilità, which provides updated information regarding parking prices, subscriptions, regulated parking zones, and parking structures operating within the city of Brescia.

Additional information was obtained from municipal tariff regulations issued by the Municipality of Brescia, which define maximum parking fees, resident discounts, subscription schemes, and differentiated parking conditions by user category.

As a benchmark for calibration, the tariff values were also compared with mobility and parking studies published by ISFORT and the Italian Ministry of Infrastructure and Transport regarding typical parking costs in medium-sized Northern Italian cities.

The tariff structure was developed separately for the three user categories considered in the model. The estimation process started from the actual hourly parking prices observed in the city. In particular:

- central regulated areas (Zona A) charge approximately €2.40/hour,
- semi-central areas (Zona B) approximately €1.50/hour,
- structured parking facilities such as Autosilo, Crystal, and Freccia Rossa approximately €1.70/hour during the first hours of parking,
- night tariffs are generally reduced to approximately €1.00/hour.

Special tariff schemes for residents were also incorporated into the model. Residents owning the “Park City Card” benefit from approximately 50% discounts on standard hourly tariffs, while monthly subscriptions correspond to an equivalent hourly cost between €0.18 and €0.23/hour.

Using these reference values, equivalent hourly tariffs were estimated for each user category and time period.

Residents were associated with lower daytime tariffs and very low nighttime costs due to residential subscriptions:

- €0.85/hour during morning and afternoon,
- €0.18/hour during the night period.

Workers were assigned intermediate tariffs reflecting long-duration parking and reduced daily rates:

- €1.30/hour during morning and afternoon,
- €0.50/hour during nighttime.

Visitors were assigned the highest tariffs, consistently with short-term parking usage in central areas:

- €1.70/hour during the morning,
- €2.00/hour during the afternoon,
- €1.00/hour during nighttime.

Since different users occupy parking spaces for different durations, hourly tariffs alone would not accurately represent the effective economic contribution of each parking allocation.

For this reason, the hourly tariffs were multiplied by the estimated average parking duration associated with each user category:

- residents: approximately 10 hours,
- workers: approximately 7 hours,
- visitors: approximately 2.5 hours.

3.8 Distance and Unmet Demand Penalties α_k and β_k

The parameters α_k and β_k were introduced in the model in order to capture the behavioural and social costs associated with parking accessibility.

More specifically:

- α_k represents the penalty associated with walking distance for user category k ,
- β_k represents the penalty associated with unmet parking demand, that is, users who are unable to find a parking space.

These parameters play a fundamental role in the optimisation model because they allow the objective functions to incorporate not only economic costs, but also user discomfort and accessibility considerations.

The estimation of the distance penalties started from the concept of value of walking time commonly used in transportation economics literature.

European mobility studies estimate the value of pedestrian walking time for medium-sized Italian cities between €8 and €12 per hour. Using an intermediate reference value of €10/hour and assuming an average walking speed of approximately 4 km/h (around 67 metres per minute), the corresponding walking cost can be converted into a monetary value per metre:

$$\frac{10 \text{ €/hour}}{60 \text{ min/hour} \times 67 \text{ m/min}} \approx 0.0025 \text{ €/m}$$

This value was adopted as the baseline distance sensitivity coefficient.

The coefficients were then differentiated according to user behaviour and parking preferences.

Residents were assumed to be the most sensitive to walking distance, especially during nighttime periods or when returning home with groceries, children, or personal belongings. Workers were assumed to have an intermediate sensitivity, since daily commuting routines generally increase tolerance toward repeated walking distances. Visitors were considered the most flexible category, as walking is often part of the leisure or tourism experience itself.

The final coefficients used in the model are therefore:

- residents: $\alpha_{res} = 0.0040 \text{ €/m}$,
- workers: $\alpha_{work} = 0.0025 \text{ €/m}$,
- visitors: $\alpha_{vis} = 0.0015 \text{ €/m}$.

These values imply that assigning users to distant parking facilities generates higher social costs for residents than for visitors, influencing the allocation decisions produced by the optimisation model.

The unmet demand penalties β_k represent the social and behavioural cost associated with users who fail to find a parking space.

The calibration of these parameters follows an important modelling principle: the penalty associated with unmet demand must be significantly higher than the distance penalty. Otherwise, the optimisation model

would systematically prefer leaving demand unsatisfied rather than assigning users to more distant parking lots, producing unrealistic behavioural outcomes.

For this reason, the calibration was based on the following practical relationship:

$$\beta_k \geq \alpha_k \times d_k^{max}$$

where d_k^{max} represents the maximum acceptable walking distance for user category k .

For example, residents were assumed to tolerate a maximum walking distance of approximately 600 metres. Given the calibrated value of $\alpha_{res} = 0.0040$ €/m, the minimum consistent unmet demand penalty becomes:

$$0.0040 \times 600 = 2.40 \text{ €}$$

An additional discomfort component was then added in order to account for the inconvenience of searching for alternative parking solutions or failing to park near the destination.

The final unmet demand penalties adopted in the model are:

- residents: $\beta_{res} = 3.50$ €/vehicle,
- workers: $\beta_{work} = 2.50$ €/vehicle,
- visitors: $\beta_{vis} = 1.20$ €/vehicle.

These values reflect the fact that failing to find parking near home represents the highest level of inconvenience, while visitors are generally more flexible and may more easily modify their travel plans or choose alternative transportation options.

4 Formulation of the mathematical model

4.1 Operational vs Strategic Approach

The model adopts a two-level hybrid approach:

- 1) The strategic (long-term) planning decision is represented by the choice of which sites x_i to open
- 2) The daily operational efficiency is governed by the allocation of flows y_{ijkt} and by the unmet demand u_{jkt}

4.2 Decision Variable

- $x_i \in \{0, 1\}$: Binary variable that assumes value 1 if the new site i is opened, and 0 otherwise (this variable is set to 1 for the existing facilities).
- $y_{ijkt} \in [0, 1]$: Fraction of the demand of node j , user k , time t allocated to car park i .
- $u_{jkt} \in [0, 1]$: Proportion of unmet demand at node j , class k , time t .

4.3 Scenario A - Minimising Total Network Costs

The first proposed objective function of the approach is to minimise the overall fixed costs and service penalties associated with mobility:

$$\min \sum_{i \in I} F_i * x_i + \left(\sum_{i \in I} \sum_{j \in J} \sum_{k \in K} \sum_{t \in T} \alpha_k * d_{ij} * D_{jkt} * y_{ijkt} * 365 \right) + \left(\sum_{i \in I} \sum_{j \in J} \sum_{k \in K} \sum_{t \in T} \beta_k * D_{jkt} * u_{jkt} * 365 \right)$$

To better understand this formula, it can be broken down into three main sections:

1. **Fixed activation cost** $\sum_{i \in I} F_i * x_i$, represent the one-off financial investment required to open the proposed car parks.
2. **Penalty of social disutility of distance** $\alpha_k * d_{ij} * D_{jkt} * y_{ijkt} * 365$, converts into monetary terms the inconvenience experienced by users who have to walk from car park i to the demand node j .
3. **Penalty for unmet demand** $\beta_k * D_{jkt} * u_{jkt} * 365$, represent the social cost related to the failure of the system (vehicles that cannot find a parking space, adding congestion by drivers looking for a space).

4.4 Scenario B - Maximising Social Profit

The second objective is to maximises the overall economic and social balance of the network, calculated as tariff revenue minus operating costs and parking fines:

$$\max \left(\sum_{i \in I} \sum_{j \in J} \sum_{k \in K} \sum_{t \in T} P_{kt} * D_{jkt} * y_{ijkt} * 365 \right) - \sum_{i \in I} F_i * x_i - \left(\sum_{i \in I} \sum_{j \in J} \sum_{k \in K} \sum_{t \in T} \alpha_k * d_{ij} * D_{jkt} * y_{ijkt} * 365 \right) - \left(\sum_{i \in I} \sum_{j \in J} \sum_{k \in K} \sum_{t \in T} \beta_k * D_{jkt} * u_{jkt} * 365 \right)$$

To better understand this formula, it can be broken down into three main sections:

1. **Total Revenue from Tariffs** $\sum_{i \in I} \sum_{j \in J} \sum_{k \in K} \sum_{t \in T} P_{kt} * D_{jkt} * y_{ijkt} * 365$, represents the revenue generated by users who pay to use the parking service.
2. **Fixed activation cost** $\sum_{i \in I} F_i * x_i$, represent the one-off financial investment required to open the proposed car parks.
3. **Penalty of social disutility of distance** $\sum_{i \in I} \sum_{j \in J} \sum_{k \in K} \sum_{t \in T} \alpha_k * d_{ij} * D_{jkt} * y_{ijkt} * 365$, converts into monetary terms the inconvenience experienced by users who have to walk from car park i to the demand node j .
4. **Penalty for unmet demand** $\sum_{i \in I} \sum_{j \in J} \sum_{k \in K} \sum_{t \in T} \beta_k * D_{jkt} * u_{jkt} * 365$, represent the social cost related to the failure of the system (vehicles that cannot find a parking space, adding congestion by drivers looking for a space).

While scenario A describes a conservative approach, the system has no incentive to invest other than to avoid the penalties associated with unmet demand, the second scenario B describes a proactive approach. Thanks to the factor of the revenue P_{kt} , the mathematical algorithm assesses whether the opening of a new car park is likely to attract enough demand to generate revenue that exceeds the annual fixed cost of opening it.

4.5 Model Constraints

1. Demand Satisfaction

$$\sum_{i \in I} y_{ijkt} + u_{jkt} = 1 \quad \forall j \in J, \forall k \in K, \forall t \in T$$

This constraint ensures that the model captures the entire demand for every demand node j (e.g. Spedali Civili), for every user type k (e.g. workers) and in every time slot t (e.g. morning). The sum of the fractions of cars allocated to the various car parks, combined with the fraction of cars that cannot find a space (unsatisfied demand), must be exactly 100% (i.e. 1).

- $\sum_{i \in I} y_{ijkt}$: represents the sum of the fractions of demand that the model decides to allocate across all available parking spaces i .
- u_{jkt} : represents the fraction of the demand that does not find a space in any car park.
- $= 1$: ensures that no car “disappears” into thin air. If demand is not parked, the model is forced to activate the penalty variable u to make up the difference and reach 1.

2. Parking Capacity

$$\sum_{j \in J} \sum_{k \in K} D_{jkt} * y_{ijkt} \leq C_i * x_i \quad \forall i \in I, \forall t \in T$$

This constraint requires compliance with the physical limit of available parking spaces in each individual facility i and for each time slot t .

- $\sum_{j \in J} \sum_{k \in K} D_{jkt} * y_{ijkt}$: calculate the total number of actual vehicles that, from all nodes j and for all categories k , are directed to car park i during time slot t .
- C_i : is the maximum number of parking spaces in car park i .
- x_i : is the binary decision variable (0 or 1).

3. Non-negativity and Binary

$$y_{ijkt} \geq 0 \quad \forall i, j, k, t$$

$$u_{jkt} \geq 0 \quad \forall j, k, t$$

$$x_i \in \{0, 1\} \quad \forall i \in I$$

4.6 Classification of the problem

4.6.1 MILP, CFLP

The model is structured as a Capacitated Facility Location Problem (CFLP), i.e. a facility location problem with capacity constraints. In its standard form, the CFLP aims to determine which facilities to open (a binary choice) and how to allocate customer demand across those facilities, without exceeding the maximum capacity of each facility and whilst minimising the sum of fixed and variable costs. Since the model includes both integer and binary decision variables ($x_i \in \{0, 1\}$ for site opening) and continuous real variables (y_{ijkt} , $u_{jkt} \in [0, 1]$ for flow fractions and unmet demand), it falls within the class of Mixed-Integer Linear Programming (MILP) models.

4.6.2 Dimensionality of the model

To assess the complexity of the model, it is possible to calculate the exact number of variables and constraints generated by the model's algebraic equations.

Calculation of variables

- Number of parking spaces (existing + proposed) : $|I| = 31$.
- Number of demand nodes : $|J| = 20$.
- Number of user types : $|K| = 3$.
- Number of time slots : $|T| = 3$.
- Binary Variable (x_t): 31
- Continuous flow variables (y_{ijkt}): $|I| * |J| * |K| * |T| = 31 * 20 * 3 * 3 = 5.580$ variables.
- Continuous variables for unmet parking needs (u_{jkt}): $|J| * |K| * |T| = 20 * 3 * 3 = 180$ variables.
- Total variables: $31+5.580+180=5.791$ variables.

Calculation of constraints

- Constraints on Demand Satisfaction : One constraint for each combination of node, user and time slot: $|J| * |K| * |T| = 20 * 3 * 3 = 180$ constraints.
- Constraints on Parking Capacity : One parking restriction for each parking space during each time slot: $|I| * |T| = 31 * 3 = 93$ constraints.
- Total Constraints (excluding non-negativity constraints): $180+93=273$ constraints.

4.6.3 Selected Solver

For the mathematical solution of the scenarios has been adopted the GLPK (GNU Linear Programming Kit) solver, called within the Python environment via the Pyomo modelling interface.

5. Python Implementation

5.1 Libraries and software environment

The implementation relies on different Python libraries, among which Pyomo with the open-source solver GLPK, Numpy for numerical operations on parameter arrays, Pandas to handle tabular data and inspecting parameter dictionaries, and math as a standard library module used for the distance computation. The core dependency is Pyomo, which provides a high-level algebraic modelling language for mathematical programming problems. Pyomo cleanly separates the model structure from the data and from the solver, making it straightforward to switch between the two objective functions without rewriting any constraints. GLPK is installed as a system-level executable and invoked by Pyomo through the SolverFactory interface; it handles the mixed-integer linear programming problem via the branch-and-bound algorithm.

5.2 Code structure

The implementation is organised as a single Python script divided into three sequential blocks, each preceded by a header comment.

Block	Label	Contents
Block 1	Data and parameters	Sets I, J, K, T; parameters C, F, d, D, P, REVENUE, α , β ; coordinate-based distance computation via Haversine.
Block 2	Model definition	ConcreteModel instantiation; Pyomo Sets and Params; variables x, y, u; constraints demand_sat and capacity; declaration of both objective functions.
Block 3	Solving and output	Sequential activation of Scenario A and Scenario B; GLPK solver calls; extraction and printing of x, u, and Y results; comparative summary.

Table 5.2 — Structure of the Python implementation.

Block 1 defines all sets and parameters required by the model. Each parking site is stored as a tuple in the parking_data dictionary, containing its name, geographic coordinates, capacity, cost, and a Boolean flag indicating whether it is an existing facility or a candidate.

```
parking_data = {
    "I01": ("Camper Poliambulanza",      45.5215, 10.2368, 10,      0, True),
    "I02": ("Crystal Superficie",       45.5358, 10.2192, 50,      0, True),
    "I03": ("Apollonio",                45.5292, 10.2182, 115,     0, True),

    "I30": ("Parcheggio ex Pietra",      45.5422, 10.2058, 450, 225000, False),
    "I31": ("Mega Park&Ride Sud",       45.5052, 10.2398, 950, 237500, False),
}

I = list(parking_data.keys())
I_exist = [i for i in I if parking_data[i][5]]
I_new = [i for i in I if not parking_data[i][5]]
```

The capacity $C(i)$ and the annual cost $F(i)$ for each parking site are derived from this dictionary too. Costs are annualised within Block 1 by dividing the total construction cost by the assumed useful life of 20 years. This ensures that $F(i)$ is expressed in €/year, the same temporal unit as the revenue and penalty terms in the objective functions (which are multiplied by 365 to convert daily flows to annual totals).

Also the J nodes are derived from a dictionary called node_data, that contains their id, name, and geographical coordinates (latitude and longitude).

```
node_data = {
    "J01": ("Piazza della Vittoria",    45.5381, 10.2192),
    "J02": ("Piazza Paolo VI / Duomo",  45.5387, 10.2213),
    "J03": ("Teatro Grande",            45.5370, 10.2175),
}
```

The distance matrix d is computed directly within Block 1 using the Haversine formula with an urban correction factor of 1.3, as described in Section 3.5. The result is stored as a Python dictionary with keys (i, j) , which Pyomo accepts natively as a parameter initialiser.

```
# d(i,j): distance matrix (from i to j)
def haversine(lat1, lon1, lat2, lon2):
    R = 6371000 # average radius of the earth in meters
    p1, p2 = math.radians(lat1), math.radians(lat2)
    dp = math.radians(lat2 - lat1)
    dl = math.radians(lon2 - lon1)
    a = math.sin(dp/2)**2 + math.cos(p1)*math.cos(p2)*math.sin(dl/2)**2
    return R * 2 * math.asin(math.sqrt(a))

urban_factor = 1.3

d = {}
for i in I:
    _, plat, plon, *_ = parking_data[i]
    for j in J:
        _, jlat, jlon = node_data[j]
        d[i, j] = round(haversine(plat, plon, jlat, jlon) * urban_factor)
```

The demand parameter $D[j, k, t]$ is built from three raw inputs per node — residential population, number of workers, and daily visitors — through the application of the conversion coefficients described in Section 3.6. The time distribution is applied as a final step, disaggregating the daily type-level demand into the three time periods.

```
D = {}
for j in J:
    res, wor, vis = demand_raw[j]
    d_res = round(res * tax_mot * use_rate)
    d_wor = round(wor * work_car)
    d_vis = round(vis * vis_car)
    base = {"res": d_res, "wor": d_wor, "vis": d_vis}
    for k in K:
        for ti, t in enumerate(T):
            D[j, k, t] = round(base[k] * time_dist[k][ti])
```

Block 2 instantiates a Pyomo ConcreteModel and populates it with sets, parameters, variables, constraints, and objective functions. The binary variable $x(i)$ is declared over the full set I . For existing sites, it is immediately fixed to 1 using Pyomo's `.fix()` method, which removes these variables from the solver's decision space while retaining them in the constraint expressions. Only the five candidate sites remain as free binary variables.

The allocation variable $y(i, j, k, t)$ is declared as a continuous variable bounded in $[0, 1]$, representing the fraction of demand $D(j, k, t)$ assigned to site i . The unmet demand variable $u(j, k, t)$ is also continuous and bounded in $[0, 1]$, representing the fraction of demand that is not served by any parking lot.

Two constraints are imposed. The demand satisfaction constraint ensures that, for every combination of node, user type, and time period, the sum of all allocated fractions plus the unmet fraction equals exactly 1, so that the entire demand is accounted for:

```
def demand_satisfaction(model, j, k, t):
    return (sum(model.y[i,j,k,t] for i in model.I)
            + model.u[j,k,t]
            == 1)

model.demand_sat = pyo.Constraint(
    model.J, model.K, model.T,
    rule=demand_satisfaction
)
```

The capacity constraint ensures that the total number of vehicles assigned to parking lot i in time period t does not exceed its capacity. Since y is expressed as a fraction, the demand $D(j, k, t)$ is used as a multiplier to convert fractions into absolute vehicle counts before comparing with the capacity:

```
def capacity_constraint(model, i, t):
    return (sum(model.D[j,k,t] * model.y[i,j,k,t]
                for j in model.J
                for k in model.K)
            <= model.C[i] * model.x[i])

model.capacity = pyo.Constraint(
    model.I, model.T,
    rule=capacity_constraint
)
```

Both objective functions are declared as Pyomo Objective components. Objective B is immediately deactivated after declaration so that the model is ready for the first solve (Scenario A). The sense attribute controls the optimization direction: `pyo.minimize` for Scenario A and `pyo.maximize` for Scenario B.

Block 3 solves the model under both objective functions and extracts the key results for each scenario. The solver is instantiated once and reused for both runs. The two scenarios are solved sequentially using the deactivate / activate pattern. After the first solve, `obj_A` is deactivated and `obj_B` is activated before calling the solver again. Because the constraint set does not change between the two runs, Pyomo does not need to rebuild the model — only the objective expression is swapped.

```
model.obj_A = pyo.Objective(rule=obj_cost_min,      sense=pyo.minimize)
model.obj_B = pyo.Objective(rule=obj_social_profit, sense=pyo.maximize)
model.obj_B.deactivate()

solver = pyo.SolverFactory('glpk')

results_A = {}
results_B = {}
```

After each solve, the values of all decision variables are extracted using `pyo.value()` and stored in dictionaries. Three categories of results are collected: the opening decisions $x(i)$ for candidate sites, the allocation fractions $y(i, j, k, t)$ for all pairs, and the unmet demand fractions $u(j, k, t)$. The termination condition is checked before extracting values to ensure the solver reached an optimal solution.

The computational cost of both solves is negligible given the problem size. The model contains 5,765 variables (5,580 continuous y , 5 binary x , 180 continuous u) and 273 constraints (180 demand satisfaction, 93 capacity). On a standard laptop, GLPK resolves each scenario in under two seconds, confirming that the problem is well within the tractable range for this class of solver.

6. Results and Comparative Analysis

6.1 Scenario A Results — Cost Minimisation

Scenario A represents the baseline optimisation problem, where the objective function minimises the total system cost, including parking opening costs, walking distance penalties, and unmet demand penalties.

The total estimated parking demand generated by the urban system is approximately 53,914 vehicles per day across all time periods, while the total parking capacity currently available is 14,189 parking spaces. This imbalance already suggests the presence of structural parking pressure within the city.

Under Scenario A, the solver identifies an optimal annual cost of approximately €31.4 million. In order to minimise total system costs, only two candidate parking facilities are activated:

- I27 — HUB Lamarmora Sud,
- I29 — Ospedale Nord-Ovest.

The remaining three candidate sites are not opened because their construction costs are not sufficiently compensated by the reduction in unmet demand and walking penalties.

The resulting allocation prioritises the most strategically efficient parking expansions, particularly near hospital and mixed residential–tertiary demand zones.

However, the cost-minimisation approach leaves a substantial portion of parking demand unsatisfied:

- residents: 5,610 vehicles/day (37.2% unmet),
- workers: 12,660 vehicles/day (44.4% unmet),
- visitors: 8,369 vehicles/day (81.4% unmet).

The particularly high unmet demand among visitors suggests that, under a purely cost-oriented strategy, the model tends to sacrifice low-priority and lower-penalty users in order to reduce infrastructure costs.

In terms of spatial allocation, the average walking distance for served users remains relatively low at approximately 293 metres, indicating that the model strongly prefers short-distance assignments rather than expanding parking availability through additional facilities.

6.2 Scenario B Results — Social Profit Maximisation

Scenario B represents the main analytical configuration of the project. In this case, the objective function maximises social profit by balancing tariff revenues, parking activation costs, walking distance penalties, and unmet demand penalties.

The optimisation produces a significantly different solution compared to Scenario A. The optimal objective value reaches approximately €51.4 million annually, and all five candidate parking facilities are activated:

- I27 — HUB Lamarmora Sud,
- I28 — Parcheggio Sud Stazione,
- I29 — Ospedale Nord-Ovest,
- I30 — Parcheggio ex Pietra,
- I31 — Mega Park&Ride Sud.

Unlike the baseline configuration, the social-profit approach recognises that additional parking infrastructure can generate both direct economic revenues and indirect accessibility benefits.

The activation of all candidate facilities substantially reduces unmet parking demand:

- residents: 6,863 vehicles/day (45.5% unmet),
- workers: 6,019 vehicles/day (21.1% unmet),
- visitors: 5,130 vehicles/day (49.9% unmet).

The most significant improvement concerns workers, whose unmet demand decreases dramatically compared to Scenario A. Visitor accessibility also improves considerably, although visitors still experience the highest unmet demand percentage overall due to the strong concentration of tourism demand in central areas.

The model achieves these improvements by accepting higher average walking distances. Under Scenario B, the average served distance increases to approximately 628 metres. This reflects a more extensive use of peripheral and park-and-ride facilities in exchange for greater parking coverage and lower unmet demand.

From an operational perspective, the optimisation demonstrates that a revenue-oriented urban parking system tends to favour broader infrastructure utilisation and larger parking networks.

6.3 Comparative Analysis: Scenario A vs Scenario B

The comparison between the two optimisation scenarios represents one of the most relevant analytical contributions of the project.

The results clearly show that cost efficiency and social welfare do not coincide.

Scenario A adopts a conservative infrastructure strategy, with only 2 new parking facilities being opened, average walking distances that remain low, while almost half of the total parking demand remains unsatisfied.

Scenario B instead adopts an expansion-oriented strategy; all 5 candidate sites are activated, unmet demand decreases significantly, but users are willing to walk longer distances on average.

Overall unmet demand decreases from:

- 26,639 vehicles/day (49.4%) in Scenario A,
to:
- 18,012 vehicles/day (33.4%) in Scenario B.

At the same time, average walking distance increases from:

- 293 metres,
to:
- 628 metres.

The comparison therefore reveals a fundamental trade-off within urban parking management. Minimising costs tends to preserve accessibility for a limited number of users while excluding many others, while maximising social profit expands parking coverage at the cost of longer walking distances and higher infrastructure investments.

The analysis also highlights the strategic importance of the additional candidate sites. While HUB Lamarmora Sud and Ospedale Nord-Ovest are selected under both scenarios, the other three facilities are activated only under the social-profit objective. These facilities primarily support peripheral demand redistribution and reduce congestion in the most saturated urban zones.

6.4 User-Type Analysis

The impact of the optimisation differs substantially across the three user categories.

Residents experience relatively stable accessibility across both scenarios. Although unmet demand slightly increases under Scenario B, residents still benefit from relatively short walking distances due to their high distance penalty coefficient. The model therefore prioritises assigning residents to nearby parking facilities whenever possible, particularly during nighttime periods.

Workers are the category benefiting the most from the transition to Scenario B. Unmet demand decreases from approximately 44.4% to 21.1%, largely thanks to the activation of Ospedale Nord-Ovest, HUB Lamarmora Sud, and the southern peripheral parking facilities.

The optimisation allocates workers more aggressively toward park-and-ride structures and high-capacity parking areas, accepting moderately longer walking distances in exchange for significantly higher service coverage.

Visitors remain the most penalised category in both scenarios. This result is partly intentional within the model structure: visitors have the lowest unmet demand penalty, and the lowest walking sensitivity coefficient. Consequently, the optimisation model treats visitor demand as the most flexible component of the system. However, Scenario B still produces a major improvement for visitors, reducing unmet demand from over 81% to approximately 50%. This improvement is achieved primarily through the activation of peripheral parking facilities and stronger utilisation of large-capacity parking structures such as Freccia Rossa and Mega Park&Ride Sud.

6.5 Analysis of Parking Allocation

The allocation variable $Y(i,j,k,t)$ provides important insights regarding the operational behaviour of the parking system. Several parking facilities reach full or near-full utilisation in both scenarios, including:

- Stadio Rigamonti,
- Arnaldo Park,
- Piazza Vittoria,
- Fossa Bagni,
- S. Eufemia-Buffalora.

This suggests that these facilities already represent structurally strategic parking hubs within the city. The comparison also reveals significant differences in the utilisation of large-capacity facilities.

For example:

- Freccia Rossa increases from approximately 24.5% utilisation in Scenario A to almost 70% in Scenario B,
- Casazza changes from completely unused to more than 75% utilisation,
- Mega Park&Ride Sud serves nearly 2,000 vehicles/day once activated.

The opening of the new parking sites substantially modifies the allocation flows associated with nearby demand nodes.

For example:

- the new Ospedale Nord-Ovest facility strongly redistributes demand related to the university and hospital district,
- ex Pietra significantly improves parking availability for the Carmine residential area,
- Mega Park&Ride Sud absorbs a large portion of demand generated by the San Polo district.

The analysis of average walking distance by user category further confirms the behavioural structure embedded in the model:

- residents maintain the shortest average walking distances,
- workers accept moderate increases,
- visitors absorb the largest increase in walking distance.

In particular, visitor walking distance rises from approximately 256 metres to more than 1.2 kilometres under Scenario B, demonstrating how the model exploits visitor flexibility in order to improve global parking system efficiency.

Overall, the allocation analysis confirms that the optimisation model is capable of reproducing realistic urban parking behaviours while simultaneously identifying strategic infrastructure interventions capable of improving parking accessibility at city scale.

7. Sensitivity analysis

A sensitivity analysis was conducted in order to evaluate the impact of increasing visitor parking tariffs by €1 across all time periods. The purpose of this analysis was to assess how tariff adjustments influence the optimal parking allocation strategy, infrastructure activation decisions, and overall system performance.

The increase in visitor tariffs produced no changes in Scenario A. The annual objective value remained equal to €31.3 million, and the model continued to open only two new parking facilities: HUB Lamarmora Sud (I27) and Ospedale Nord-Ovest (I29).

Similarly, unmet demand levels for residents, workers, and visitors remained unchanged. Visitor unmet demand continued to be extremely high at 81.4%, while the average travel distance also remained stable at 296 meters.

This result is consistent with the structure of the cost-minimization objective function. Since Scenario A does not explicitly include parking revenues in the optimization criterion, modifications in tariffs do not affect the model's decisions. The optimization process remains entirely driven by infrastructure costs, unmet demand penalties, and distance penalties.

Therefore, the sensitivity analysis demonstrates that tariff policies have no influence under a purely cost-oriented planning strategy.

In contrast, the increase in visitor tariffs had a substantial effect under Scenario B, where revenues directly contribute to the objective function.

The annual social profit increased from approximately €55.9 million to €61.8 million, corresponding to an increase of nearly €5.9 million per year. This demonstrates that visitor parking revenues represent a significant source of economic value within the urban parking system.

Interestingly, the set of activated parking facilities did not change. The model continued to open the same five parking sites: I27, I28, I29, I30, and I31. This suggests that the existing infrastructure expansion strategy was already optimal from a spatial and operational perspective, even before the tariff increase.

However, the allocation of demand changed significantly. Visitor unmet demand decreased from 43.6% to 34.4%, indicating that the model allocated a larger share of parking capacity to visitors after the tariff increase. Since visitor parking became more profitable, the optimization process prioritized serving visitor demand more intensively.

At the same time, resident unmet demand worsened considerably, increasing from 42.5% to 48.5%. Worker unmet demand remained almost unchanged, increasing marginally from 17.2% to 17.3%.

This result highlights an important redistribution effect generated by the tariff increase. Because parking capacity is limited, prioritizing higher-paying visitor users reduces the availability of spaces for lower-revenue user categories, particularly residents.

The average travel distance also increased slightly from 663 meters to 684 meters. This indicates that the model relied more heavily on peripheral parking facilities in order to maximize total revenues from visitor traffic.

The sensitivity analysis reveals a strong dependence of the social-profit-maximization strategy on tariff structures. Increasing visitor tariffs significantly improves economic performance and visitor demand satisfaction, but it simultaneously generates a deterioration in equity outcomes for residents.

From a managerial perspective, this result illustrates the classical trade-off between profitability and accessibility. Higher visitor tariffs incentivize the system to allocate scarce parking resources toward the most profitable users, potentially reducing service quality for local residents.

The findings therefore suggest that tariff increases can be an effective policy instrument for improving financial sustainability and tourism-related accessibility, but they should be implemented carefully in order to avoid excessive exclusion of resident users.

Overall, the analysis confirms that pricing policies become highly influential only when revenues are explicitly integrated into the optimization objective. Under a purely cost-minimization framework, tariff modifications have no operational impact.

8. Conclusions

8.1 Research Questions and Main Findings

The objective of the SmartPark Optimization project was to investigate whether an optimization-based approach could improve the allocation of urban parking demand in the city of Brescia while balancing economic efficiency, accessibility, and social welfare.

The comparison between the two scenarios highlights an important managerial and urban planning insight: the configuration that minimises direct costs does not necessarily correspond to the configuration that maximises urban welfare.

The results also reveal the strategic importance of several new candidate facilities. In particular:

- I27 HUB Lamarmora Sud,
- I29 Ospedale Nord-Ovest,
- and I30 Parcheggio ex Pietra

emerge as highly valuable interventions due to their ability to reduce congestion in underserved residential, hospital, and mixed-use urban areas.

Another important finding concerns the behavioural structure of the model. The differentiation between residents, workers, and visitors generates significantly different allocation patterns and accessibility outcomes. Residents are consistently prioritised in terms of proximity, while visitors absorb a larger share of long-distance assignments due to their greater behavioural flexibility.

Overall, the optimisation results confirm that parking management should not be interpreted exclusively as an infrastructure problem, but rather as a behavioural and accessibility allocation problem involving heterogeneous urban users with different mobility needs.

8.2 Practical Recommendation for the Municipality of Brescia

From a practical policy perspective, the analysis suggests that neither Scenario A nor Scenario B should be implemented in a purely extreme form.

Scenario A excessively prioritises cost containment at the expense of accessibility and service coverage. Conversely, Scenario B requires substantial infrastructure investment and generates significantly higher average walking distances, particularly for visitors. For this reason, the most realistic and balanced strategy for the Municipality of Brescia would likely correspond to a hybrid intermediate scenario.

Based on the optimisation results, the recommended configuration would involve the activation of:

- I27 — HUB Lamarmora Sud,
- I29 — Ospedale Nord-Ovest,
- I30 — Parcheggio ex Pietra.

These three facilities appear to generate the highest strategic value in terms of demand coverage, reduction of congestion pressure, support for residential and hospital areas, and integration with existing mobility infrastructure.

In particular:

- HUB Lamarmora Sud improves accessibility in the Brescia Due and Lamarmora district,
- Ospedale Nord-Ovest supports both the hospital district and the university engineering campus,
- ex Pietra reduces pressure in the Carmine and southern central residential areas.

The results also suggest that peripheral park-and-ride systems can become effective only when integrated with public transportation networks and supported by pricing policies capable of encouraging modal shifts away from the private car.

More generally, the project demonstrates how optimisation tools can support evidence-based urban planning decisions by providing quantitative evaluations of alternative infrastructure strategies.

8.3 Model Limitations and Future Developments

Despite the robustness of the optimisation framework, the project presents several limitations that should be acknowledged.

First, parking demand values were estimated through indirect approximation methods due to the absence of complete neighbourhood-level parking datasets for the city of Brescia. Although the calibration process relied on official demographic, mobility, and tourism data, the resulting demand matrix remains partially dependent on behavioural assumptions and estimated coefficients.

Second, the model adopts a static formulation. Demand, parking occupancy, and traffic conditions are assumed to remain constant within each time period. In reality, urban parking systems are highly dynamic and evolve continuously throughout the day.

Third, the model does not explicitly incorporate real-time traffic congestion, public transport substitution effects, electric vehicle infrastructure, illegal parking behaviour or stochastic demand uncertainty.

Another important limitation concerns the spatial representation of accessibility. Distances were approximated using corrected geodesic calculations rather than actual pedestrian network routing. While the adopted correction factor improves realism, it cannot fully reproduce the complexity of real urban walking paths.

Future developments of the project could therefore include:

- dynamic multi-period optimisation,
- integration with GIS-based pedestrian networks,
- stochastic demand modelling,
- real-time parking occupancy prediction,
- and multimodal mobility integration with metro and bus systems.

Nevertheless, despite these limitations, the SmartPark Optimization project demonstrates that mixed-integer optimisation models can provide meaningful strategic insights for urban parking management and smart mobility planning. The framework developed in this study offers a realistic and adaptable decision-support tool capable of supporting future parking and mobility policies within medium-sized European cities such as Brescia.

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